

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in Uttarakhand, India

An Efficient DDoS Attacks Detection And Mitigation Approach In SDN-IoT Network

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Abstract

The Internet of Things (IoT) has revolutionized modern life by introducing advanced solutions and enhancing processing power. However, managing IoT networks can be challenging due to the multitude of devices and communication protocols involved. Software-Defined Networking (SDN) offers a potential solution by simplifying network management, providing network abstraction, facilitating network development, and potentially addressing the complexities of IoT networks. Despite the numerous benefits, it is important to note that IoT systems still face significant security concerns. This study uses entropy as a metric on the SDN controller in an IoT network to assess the homogeneity of network traffic. And on the SDN controller network log data is present so to filter network traffic, the technique additionally incorporates machine learning (ML) classifiers. We tested the suggested methodology's effectiveness using real-world network traffic information. SDN-IoT networks and ML are used to improve the process of detection and mitigation. The proposed method has a detection rate between 98% and 100%, demonstrating its exceptional performance. Compared to existing methods, our approach not only detects Distributed Denial of Service (DDoS) attacks with remarkable accuracy, but also enables early detection while maintaining a remarkable false-positive rate. This combination of high precision and early detection enables us to effectively identify and mitigate DDoS attacks, thereby enhancing security and minimizing the potential harm caused by malicious activities.

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Peer-review under responsibility of the scientific committee of the Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06)

Keywords: Distributed denial-of-service; Internet of Things; Software-defined networks; Entropy; Machine Learning

1. Introduction

The Internet of Things (IoT) has acquired prominence in a variety of fields, such as intelligent healthcare, intelligent urban environments, connected residences, efficient energy infrastructures, advanced transportation systems, and enhanced retail experiences. This pervasive adoption illustrates the evolution of technology that bridges the digital and physical domains [1,2]. IoT services have had a big effect on people's lives, especially those with disabilities, who can use people-centered solutions like IoT help to become more independent and take part in their communities [3]. IoT technologies have also helped make jobs less risky, like in mines, where self-driving tools keep workers away from dangerous places [4]. Environmental disasters can also be prevented with the help of IoT monitors that measure temperature, dangerous gases, and smoke. Researchers have looked at a lot of other case studies in the journals [5]. Because smart devices for consumers and businesses are getting smarter, the IoT market is expected to be worth trillions of dollars per year in the near future. This is because there is a growing demand for these devices, which can improve processes and services by analyzing data.

Even though there are many benefits to IoT, its broad use has led to more security risks and worries [6]. The variety of devices, network protocols, and communication technologies has made the system more vulnerable overall. Cyber threats can reach 70% of internet-connected devices [7]. DDoS attacks on IoT networks are different from other IoT threats in a number of ways. It can involve a lot of gadgets that have been hacked and are able to send out a lot of traffic. This can make it harder to find and stop threats, which can lead to widespread network service problems. DDoS attacks on IoT networks can have a direct effect on the services and devices that are available on the network, which can affect business operations or even important infrastructure. DDoS attacks on IoT networks often use botnets, which are groups of devices that have been hacked and can be managed directly by the attacker. Botnets are hard to find and can be used in many different kinds of attacks. Often, IoT devices are made with insufficient security controls, which make them easy for attackers to use. DDoS attacks are a big problem for the Internet of Things because IoT devices are often linked to the Internet and may not have enough protection. The goal of a DDoS attack is to stop a network resource, like a computer or device connected to the Internet of Things (IoT), from working normally. This is done by flooding the system with traffic from fake requests, which keeps genuine traffic from getting through. For example, someone could send bad code to a camera or home control device that is connected to the Internet of Things [12].

DDoS attacks are one common way that IoT settings are attacked, and they cause devices to break and data to leak [8]. Software-Defined Networking (SDN) has become a strong way to solve security problems in IoT settings by making a network that is flexible, reliable, and safe [9-11]. Experts and academics suggest putting IoT and SDN together, which would lead to the creation of an SDN-IoT platform (software Defined Network-Internet of Things). SDN has a number of features and its design is different from standard network structures in many ways. The features of SDN help to quickly and effectively secure networks while making them more flexible. Even though many methods and techniques have been suggested to find and stop DDoS attacks in IoT, they still pose a serious danger to IoT applications. This paper suggests a full security plan that takes the SDN-IoT world into account. The ML model looked at the traffic and told the mitigating module about the bad traffic to stop future attacks.

The following is a summary of this paper's contributions: The following is a summary of this paper's contributions:

- The study of DDoS attacks in SDN-IoT as well as how SDN can be used to reduce the effects of DDoS on IoT.
- A look at how some cutting-edge tools and methods can be used to find and stop problems. We paid close attention to how machine learning affects this area.
- A plan for a complete approach to protect the Internet of Things from DDoS threats.
- The development and review of the ML model that helps the SDN manager find and spot harmful data in IoT networks.
- A proposed approach for DDoS mitigation using Hashing.

This paper's subsequent sections are organized as follows: The second section examines the studies related to the detection and mitigation of DDoS attacks in diverse environments, with a particular emphasis on SDN-IoT networks. In Section 3, a proposed method is presented for detecting and mitigating DDoS attacks on SDN-IoT networks. In addition, Section 4 describes the experimental setup and demonstrates the results obtained by applying various methodologies to the dataset. Section 5 concludes with a summary of the conclusion derived from the research findings and recommendations for future research.

Literature Survey

Gadze et al. [9] introduced SDN as a novel paradigm that revolutionises the concept of software-driven networks. Nevertheless, SDN is vulnerable to DDoS attacks and other security dangers. This study concentrates on utilising deep learning models, specifically Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN), to detect and mitigate DDoS attacks within the SDN environment. The accuracy of these models was determined by comparing their accurate and inaccurate predictions. With an accuracy rate of 89.63%, the RNN LSTM model demonstrated promising performance as a deep learning approach to identify and prevent DDoS attacks in the SDN controller.

Mishra et al. [10] conducted a study that analyzed prospective attacks on various network types and devised defence mechanisms against them. Individuals who study vulnerabilities and strive to secure IoT networks can gain valuable insights from this research. Using simulated message queuing telemetry transport (MQTT) on a virtual network, the study utilized DDoS attacks, machine learning algorithms, and an artificial neural network to identify anomalies in IoT networks. A dataset containing 34 features, and eight categories of network traffic was utilized by the proposed method. The Random Forest (RF) classification algorithm demonstrated exceptional performance, attaining an accuracy rate of 99.94%. Implementing a Snort-based attack detection system further demonstrated the practicability and efficacy of the proposed method, validating its utility and feasibility.

Banitalebi et al. [11] addressed the attack posed by DDoS attacks to the security of software-defined networks (SDN). Existing methods for detecting DDoS attacks have limitations, including their reliance on network architecture, inability to detect all forms of DDoS attacks, use of obsolete and inaccurate datasets, and reliance on costly and powerful hardware infrastructure. To surmount these obstacles, the researchers proposed a novel SDN approach with three components: data acquisition, an entropy-based method, and a classification method. Experimental evaluations using the UNB-ISCX, CTU-13, and ISOT datasets demonstrated the superior efficacy of this method in detecting DDoS attacks in SDN in comparison to competing approaches.

Yang et al. [13] introduced the ADAM (Adaptive DDoS Attack Mitigation) method, which employs information entropy and automated anomaly detection techniques to effectively combat DDoS attacks. The ADAM method exhibits an impressive average detection rate of 99.13% even during intense DDoS attacks. Additionally, it significantly reduces the number of false positives by 35% to 58%, enhancing the accuracy of the detection system. And Mahmoud et al. [14] looked at all parts of DDoS attacks in SDN networks by using Information Gain (IG) and Random Forest (RF), two popular ways to choose features. To solve the problem, they have come up with a DL approach that uses LSTM and an auto encoder. Based on what they found in their research, the DL method can be used to find DDoS attacks without changing how well the device works.

Sudar et al. [15] claimed that SDN is vulnerable to DDoS attacks and suggested using verity machine learning to spot rogue traffic. Their results showed that the Decision Tree and Support Vector Machine (SVM) methods are better at finding things and getting things right. Mahmoud et al. [14] conducted research on DDoS attacks in SDN networks, using Information Gain (IG) and Random Forest (RF) for feature selection. They proposed a deep learning (DL) approach using LSTM and an autoencoder, which proved effective in detecting DDoS attacks without impacting device performance. Sudar et al. [15] highlighted the vulnerability of SDN to DDoS attacks and suggested the use of various machine learning techniques to identify malicious traffic. Their results indicated that Decision Tree and Support Vector Machine (SVM) methods outperformed others in terms of accuracy and effectiveness.

Malik et al. [16] focused on developing intrusion detection systems (IDS) for IoT networks to detect DDoS attacks. They employed feature enrichment and machine learning models, demonstrating that significant feature reduction improved the efficacy of ML-based IDS in detecting DDoS attacks in standardized IoT networks. Almaraz et al. [17] developed an intrusion detection system using machine learning and deep learning models to identify DDoS and DoS attacks on IoT networks. Their system achieved high accuracy (>99%), with Decision Tree and Multi-layer Perceptron methods being the most effective. Saghezchi et al. [18] explored the application of machine learning in detecting network anomalies in Cyber-Physical Production Systems (CPPSs) in the semiconductor manufacturing industry. They compared various supervised, unsupervised, and semi-supervised algorithms, concluding that supervised algorithms performed better in terms of detection performance.

2. Proposed Work

This study proposes a method for detecting DDoS attacks within an SDN-IoT network by combining a classification algorithm with an entropy-based technique. Detection, which includes the entropy and classification sections, and Mitigation, which entails identifying and mitigating DDoS attacks within the POX controller, are two essential components of this method.

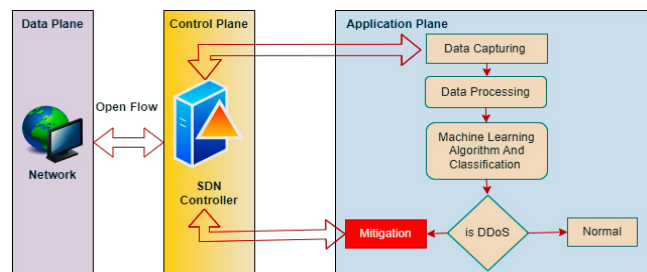


Fig. 1. Proposed work architecture of SDN-IoT

The proposed work design is made up of the following modules: Data Plane, Control Plane, Application Plane, Data Capturing, Data Processing, Machine Learning, and Mitigation Module. In the data plan module, topological setups are used to link IoT devices together. The control plane is the part of the SDN that knows what to do. All business and security apps belong to the application layer. The flow tables of the network switches are used by the data capture module to get information about IP patterns and other features. The machine learning tool looks at network data in order to find DDoS attacks. If a DDoS attack is found, the protection tool takes steps to limit the damage. Both finding and stopping DDoS attacks are parts of the application plane.

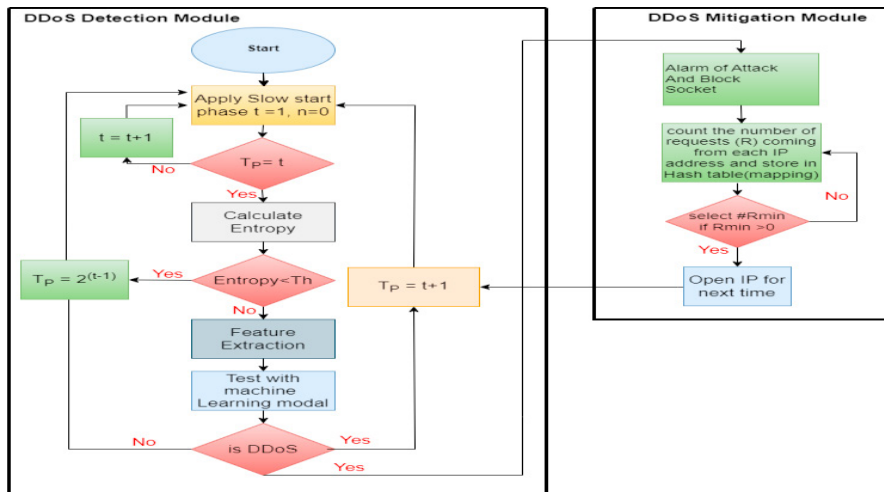


Fig. 2. Flow diagram of proposed work

2.1. DDoS Detection module

There are four steps to finding a DDoS in the detection section of the SDN controller. The work that is being presented uses a dynamic time quantum, so the detecting tool has to look at a dynamic time quantum number. The Time Period (T_p) variable starts out with a value of 5 ms. And if, the condition ($T_p = t$) is not true, the numbers of t are raised linearly. If the condition is true, the flow moves to the next step. Firstly it detect DDoS using entropy value and if DDoS is detected then ML classification algorithms used to finds a DDoS, flow is directed to the mitigation module if the ML module confirms a DDoS attack. If the attack is not confirmed, the flow is redirected to the delayed start phase, which assigns a new T_p number. The mitigation module notifies the controller of the DDoS attack by transmitting an attack message, compelling the controller to block the IP address associated with the excessive request volume.

2.2. Classification through machine learning

The Classification section is the last part of the third module. It talks about how traffic flows are put into different groups based on their traits, network conditions, and attack patterns. Because it is more flexible, this method is better able to adapt to changing network conditions and attack trends. The proposed classification system consists of five interconnected modules: flow collector, machine learning algorithm, detection, prevention, and flow manager. Using the CICFlowMeter software, the flow analyzer module effectively classifies incoming flows as either suspicious or secure based on their previous treatment. The protection module modifies the flow manager to actively thwart DDoS attacks, utilising the ML method's insights for detection module decision-making.

2.3. DDoS attacks Mitigation Module

Organizations employ a variety of countermeasures against DDoS attacks, including traffic screening and rate limiting. When a DDoS is found by the machine learning method in the suggested work, the mitigation module will be turned on. And it stops rogue IP addresses, which are devices from which a lot of requests are coming and add them to a map. As IOT network has fixed devices and which has unique IP address so we can use map. A map is a key-value pair where key is IP address and value is request count (R) per IP address. System find the min value count (R_{min}) in hash table and then unblock the IP address and remove from map for next chance to communicate with server.

3. Experiment Setup and Results

So that we could measure how well our method worked, we came up with tests that used different attack situations. Using virtual situations, the suggested method for finding DDoS attacks on the SDN controller can be quickly tested. The structure of the experimental SDN testbed [21] before you can use this approach in SDN, you need to add a DDoS attack monitoring tool to the Open Flow Switch. Network configuration is needed to start SDN networks, so the proposed work uses a simulated network structure with 64 hosts generated on the Mininet network emulator and each with an IP address from 10.0.0.1 to 10.0.0.64, a POX controller and an Open-Flow [19] switch, as well as an attacker who runs all servers on the Ubuntu Linux operating system and uses the Python programming language.

3.1. Experimental Analysis

The evaluation grid was used to evaluate the performance of the proposed method across multiple metrics, including precision, memory consumption, false positive rate (FPR), and true positive rate (TPR). In this context, 'P' indicates the number of actual positive cases (attacks), whereas 'N' indicates the number of actual negative cases (non-attacks). TPR represents the number of assaults correctly identified by the detection system, whereas FPR represents the number of instances in which benign events were incorrectly identified as threats. The alert rate is determined by dividing the total number of categorised samples by the number of attacks detected. This exhaustive evaluation grid permits a complete evaluation of the method's efficacy.

3.2. Performance evaluation of multiple ML algorithms in CICFlowMeter created dataset:

The proposed work employed four machine learning algorithms, namely LR, DT, SVM, and Catboost, for the purpose of classification. Among these algorithms, SVM and Catboost demonstrated superior accuracy compared to the other classification methods. Table 1 provides a comprehensive comparison of the results achieved by the four machine learning algorithms.

Table 1. Results of proposed work for DDoS detection

Algorithm	Accuracy(%)	F1-Score (%)		Precision(%)		Recall (%)	
		0	1	0	1	0	1
LR	99.96	100	00	100	00	100	00
DT	99.98	100	47	100	44	100	50
SVM	99.99	100	00	100	00	100	00
Catboost	99.99	100	40	100	100	100	25

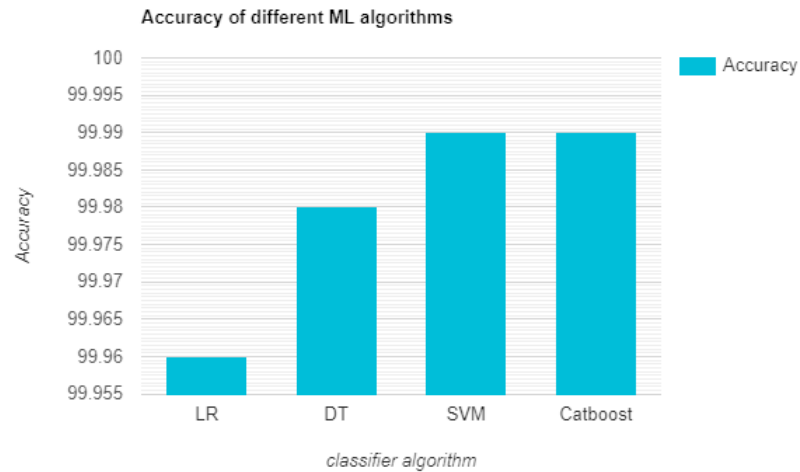


Fig.3. Accuracy of different ML algorithms using CICFlowMeter dataset

3.3. TPR and FPR values for various ML algorithms on a dataset created by CICFlowMeter

We have plotted graph between TPR and FPR presented in table 2 and as we see in case of DT and Catboost FPR rate almost minimal compare to SVM and LR.

Table 2. Results of proposed work for DDoS detection

Algorithm	TP	FP	FN	TN	TPR	FPR
LR	86552	21	8	0	0.999	1.0
DT	86568	5	4	4	0.999	0.55
SVM	86573	0	8	0	0.999	nan
Catboost	86573	0	6	2	0.999	0.00

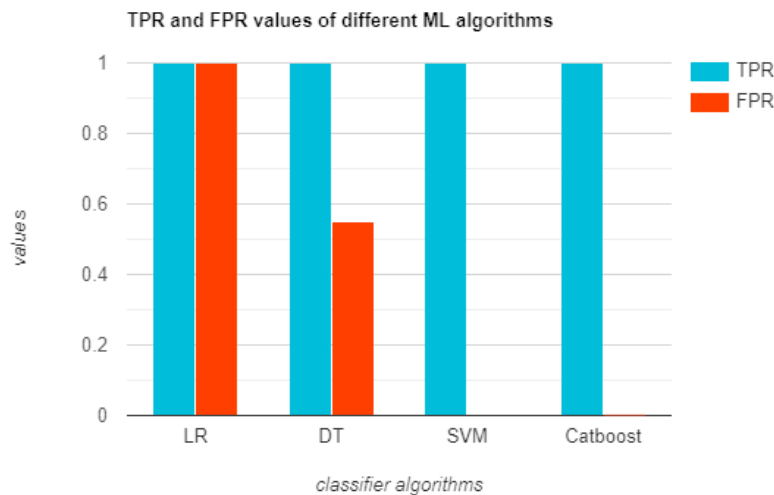


Fig.4. TPR and FPR values of different ML algorithms using CICFlowMeter dataset

3.4. Comparison of the proposed method with existing works

The proposed approach outperforms the other alternatives works significantly. Yang [13] employs information entropy and SVM-based detection, however because to the limited representation range of entropy, it is confined to recognising assaults within a certain range. Hosseini et al. [22] use a variety of classifiers, including MLP, KNN-based Divergence Test, NB, Random Forest, and, Decision Tree but they pay a price for each classifier's feature subset. Sudar [15] chooses a subset of characteristics and employs SVM and decision tree classifiers, but it performs poorly. In contrast, the suggested technique uses entropy-based detection and pick top 40 features based on correlation matrix out of 80 and leverages ML algorithms such as LR, DT, SVM, and Catboost. When compared to single classifier techniques, it outperforms them in terms of accuracy. The suggested approach achieves 99.9% accuracy, 100% precision, 98% TPR, and 1% FPR shown in Table 3. It detects both high-rate and low-rate threats successfully. In terms of accuracy and detecting capabilities, the suggested technique outperforms the mentioned alternatives.

Table 3. Results of proposed work for DDoS detection

Reference	Features	Approach	Dataset	Strength	Experiment results	Drawbacks
Yang [13]	packet types, destination port number, network packets, Source IP, destination IP, source port number,	Information entropy and SVM based	NSL KDD, DARPA KDD	Suitable for large networks	Accuracy=NA Precision=99% TPR=NA% FPR=NA	Due to the limited representation range of Entropy, it can only identify attacks that fall within a specific range.
Hosseini et al. [22]	The subset of features chosen in accordance with classifiers is referred to as the forward selected feature set.	Decision Tree, KNN-based Divergence Test and the forward-NB, Random Forest, and MLP	generated dataset, NSL KDD	Multiple classifiers are used to detect a wide variety of attacks.	Accuracy=98.9% Precision=99% TPR=87% FPR=1%	The cost of selecting a feature subset for each classifier
Sudar[15]	Select some features from dataset	DT, SVM	KDD99	DT has lower performance than SVM	Accuracy=85% Precision=80% TPR=NA FPR=NA	Performance is not upto mark
Proposed work	Choose a subset of 40 features from the original set of 80 features that are relevant to the classifiers.	entropy based detection and using ML Algorithm LR, DT, SVM, Catboost	NSL KDD, datasets and real time dataset	The detection system is capable of identifying both high-rate and low-rate attacks, making it	Accuracy=99.9% Precision=100% TPR=98% FPR=1%	Computationally complex

more
effective
than
single
classifier
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es in
terms of
accuracy.

4. Conclusion and future work

SDN has become a hopeful way to handle the traffic that IoT devices add to networks, while machine learning technology provides classification on a dataset. This paper looks at how dangerous DDoS attacks are in IoT and SDN settings and how easy they are to launch. It also looks at some of the most cutting-edge ways to stop DDoS attacks machine learning. This paper looks into new methods and techniques that can help make services for the IoT safer. In order to reach this goal, we look at the benefits of putting SDN concepts and machine learning models into IoT settings. We suggest an IoT system that is completely safe and uses SDNs with single controller and ML models to identify and mitigate DDoS attacks. Also, firewalls and security tools can be used to protect the data plane. We go into detail about the installed ML-based classification system and show how well it works. The whole applied system is putting onto mini-net simulations and using POX controller for data storages and monitor whole SDN-IoT system. proposed approach involving entropy and machine learning (ML) on various datasets, it was discovered that entropy-based methods have a remarkable detection rate spanning from 98 to 100 percent. These methods excel at both identifying DDoS attacks and mitigating their effects. Future research seeks to increase the network's efficacy by employing multiple SDN controllers, even though the paper-mentioned study used a single controller to detect DDoS attacks. In addition, the proposed study employed a static threshold for entropy, whereas future research will focus on implementing a dynamic threshold for enhanced precision.

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